MECH 328 – DYNAMICS AND VIBRATIONS

Course Syllabus – Fall 2026

This is your course syllabus. Please download the file and keep it for future reference.

TEACHING TEAM

LAND ACKNOWLEDGEMENT

Queen's University is situated on traditional Anishinaabe and Haudenosaunee Territory. See: http://www.queensu.ca/encyclopedia/t/traditional-territories

INCLUSIVITY STATEMENT

Queen’s students, faculty, and staff come from every imaginable background – small towns and suburbs, urban high rises, Indigenous communities, and from more than 100 countries around the world. You belong here: https://www.queensu.ca/vpcei/support/overview.

COURSE INSTRUCTOR

Quinn Yetman, BASc

Department of Mechanical and Materials
Engineering

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Office location: McLaughlin 215
Office hours: Available by request

MECH 328 (F 3-0-0.5 3.5)

Course Description

This course covers the kinematics and dynamics of rigid bodies in two and three dimensions, as well as an introduction to vibrations. Topics in dynamics include: mathematically rigorous kinematic analysis, Newton's laws, energy methods, impulse and momentum methods, mass moments of inertia, and gyroscopic motion. Topics in vibrations include: free and forced vibration of single-degree-of-freedom systems, undamped and damped systems.

(0/11/0/17/14)

Mode of Instruction

The delivery mode of this course is in-person.

Course Location and Time

Lecture:
Monday, 1:30pm-2:30pm – Dunning Auditorium
Friday, 11:30am-12:30pm – Dunning Auditorium

Active Learning Session:
Thursday, 11:30am-1:30pm – Mitchell 215-235 (Section 1)
Tuesday, 2:30pm-4:30pm – Mitchell 215-235 (Section 2)

Pre-Requisite Knowledge

This course is designed for learners with some background in particle mechanics, vectors, and ordinary differential equations.

Prerequisite courses: MECH 228 or MECH 229 or ENPH 225

Course Learning Outcomes (CLO)

By the end of this course, students should be able to:

CLO	Description	Indicator[1]
CLO 1	Apply derivatives of vectors, constraints, ground fixed coordinates, rotating coordinates, relative and absolute motion, Coriolis acceleration, to analyze the kinematics of rigid bodies in a plane.	[KB-ES-Classical Mechanics],[PA-Solve]
CLO 2	Apply free body diagrams, force balances, moment balance, moments of inertia, principles of work and energy, impulse and momentum to analyze the kinetics of rigid bodies in a plane.	[PA-I],[PA-F],[PA-S]

[1] For further information on graduate attributes and indicators, please click here.

CLO	Description	Indicator[1]
CLO 3	Apply 3D kinematic analysis, 3D force balances, the inertia tensor, angular momentum vectors, 3D moment balance, gyroscopic effects to determine the three dimensional dynamics of rigid bodies.	[PA-F],[PA-S]
CLO 4	Apply concepts of free vibrations, forced vibrations, damping, and energy methods to model vibration and determine time response for one degree of freedom systems.	[KB-ES],[PA-S]
CLO 5	Design a mechanism to meet specified 2D kinematics and dynamics requirements.	[PA-A],[PA-S],[TW-C],[TW-D],[TW-F],[DE-D],[DE-S],[DE-A]

Course Evaluation

Assessment Weighting

Assessment Tool	Due Date[1]	Weight[1]	Alignment with CLOs
In-Lecture Questions	**Weekly**	**5%**	**All**
Active Learning (ICAs)	**Weekly**	**20%**	**All**
Final Project	**Week 12**	**15%**	**1,2,5**
Progress Report	Week 9	5%	1,2,5
Final Report	Week 12	10%	1,2,5
Midterms	**Week 7 and Week 12**	**30%**	**1,2,3**
Midterm #1	Week 7	15%	1,2
Midterm #2	Week 12	15%	2,3
Final Exam[2]	**Please see the Final Exam schedule posted by OUR Exams Office**	**30%**	**1,2,3,4**
		100%	

[1] The evaluation weighting and due dates are subject to change given the timeline and delivery of the course.

[2] **You must pass the final exam to pass the course.**

Assessment Descriptions

In-Lecture Questions

Submit a brief answer to the question posted in the each of the lectures. Will be based on content from the active learning sessions and previous lectures (marked for completion, only need to do 20 for 100%).

Active Learning

There will be between 4 and 6 group-based in class actives (ICAs) designed to be completed by the end of the Active Learning Sessions. These will be due 3 days after the Active Learning Sessions.

Final Project

Design project based around 2D planar analysis. More details will be provided part-way through the course. A short progress update will be due in Week 9, and the final full report will be due in Week 12.

Midterms

There will be two Midterms. The date, time and location of the mid-term will be announced through onQ.

Final Exam

The data, time and location of the final examination will be announced through SOLUS.

Grading

All assessments in this course will receive numerical percentage marks. The final grade you receive for the course will be derived by converting your numerical course average to a letter grade according to the established Grade Point Index.

Students must pass the Final Exam to pass this course. If they pass the rest of the components but fail the Final Exam, an FR will be assigned, and the students can apply to write a supplemental exam.

Feedback on Assessments

The teaching team will provide feedback on graded activities. You can expect feedback on your assessments within seven days of the due date.

Accessing Your Final Grade

Your final grades will show on SOLUS. Official transcripts showing final grades will be available on the Official Grade Release Date. Please note that in official transcripts, a mark of IN (incomplete) is considered a grade, and your transcript is released with this grade.

Course Materials

Suggested Textbooks (Optional)

The Practice of Engineering Dynamics (ISBN-13: 9781119053705)

This course will follow the general outline and use content from this textbook.

Cost: $148.95

Engineering Mechanics: Dynamics (ISBN-13: 9780135426029)

Some practice problems and examples will be pulled from this textbook (practice problems will be posted on OnQ). Additional, not posted, practice problems from this textbook may be beneficial for students.

Cost: $79.99 (e-textbook)

Other Material

All other course material is accessible via OnQ.

Required Calculator

A Casio 991 is required. **ONLY** this type of non-programmable, non-communicating calculator will be allowed during tests and exams.

Suggested Time Commitment

This course represents a study period of one term spanning 12 weeks. Learners can expect to invest on average 7-9 hours per week in this course. Learners who adhere to a predetermined study schedule are more likely to successfully complete the course.

Weekly Course Learning Outcomes

Week	Learning Outcomes	Assessment
1	**Introduction & Rigid-Body Kinematics** By the end of this week, learners will be able to: • State the degrees-of-freedom of a planar mechanism using its joint constraints [CLO1] • Derive position vectors for bodies [CLO1]	
2	**Vector Derivatives & Rotating Reference Frames** By the end of this week, learners will be able to: • Rigorously derive the time-derivative of a vector expressed in a rotating reference frame [CLO1] • Derive position, velocity, and acceleration loop closure equations [CLO1] • Apply the general relative-motion equation to constrained mechanisms [CLO1]	Vectors, DOF, and Simulation ICA [CLO1]
3	**Introduction to Rigid Body Kinetics** By the end of this week, learners will be able to: • Derive $\Sigma F = ma_G$ for a rigid body from a system of particles [CLO2] • Derive the 2D moment equation $\Sigma M = I\alpha$ for a rigid body [CLO2]	
4	**Moments of Inertia & Newton's Laws Problem Procedure** By the end of this week, learners will be able to: • Compute moments of inertia using parallel axis theorem [CLO2] • Apply a systematic 4-step procedure to solve rigid body problems [CLO1] [CLO2]	Vector Derivatives and Kinetics ICA [CLO1]

Week	Learning Outcomes	Assessment
5	**Multi-Body Systems** By the end of this week, learners will be able to: • Analyze coupled multi-body systems using force and moment balances [CLO1] [CLO2]	
6	**Impulse & Momentum** By the end of this week, learners will be able to: • Compute the linear and angular momentum of a rigid body [CLO2] • Apply the impulse-momentum theorem to planar rigid body kinetics problems [CLO2] • Integrate kinematics, force/moment balances, and impulse momentum methods to solve a comprehensive multibody planar problem [CLO1] [CLO2]	Multibody Planar Motion ICA [CLO1] [CLO2]
7	**Work & Energy Methods** By the end of this week, learners will be able to: • Derive the work-energy theorem for a particle and extend it to a rigid body [CLO2] • Compute work done by a force and/or moment [CLO2] • Apply gravitational and elastic potential energy to solve rigid body energy problems [CLO2]	Midterm #1 [CLO1] [CLO2]
8	**Introduction to 3D Dynamics** By the end of this week, learners will be able to: • Recognize that planar kinematics methods extend directly to 3D [CLO1] [CLO3] • Explain why the 2D moment equation $\Sigma M = I\alpha$ does not hold for a rigid body in 3D [CLO3] • Derive the general 3D moment balance equation [CLO3]	Energy ICA [CLO2]

Week	Learning Outcomes	Assessment
9	**Inertial Tensor** By the end of this week, learners will be able to: • Derive and compute the 3D inertia tensor of a rigid body [CLO3]	Project Progress Report [CLO5]
10	**Euler's Equations, Gyroscopic Effects, & Intro to Vibrations** By the end of this week, learners will be able to: • Apply Euler's moment equations to analyze the rotation of a symmetric rigid body [CLO3] • Predict gyroscopic effects from the angular momentum vector [CLO3] • Linearize a nonlinear rigid body equation of motion about a stable equilibrium to obtain a vibration model [CLO2] [CLO4]	
11	**Free Vibration of a Single-Degree-of-Freedom System** By the end of this week, learners will be able to: • Formulate the equations of motion for an undamped free vibration of a 1 degree-of-freedom system [CLO4] • Distinguish restoring forces from constant forces in a vibrating system [CLO4] • Determine the natural frequency of a 1 degree-of-freedom system from its properties [CLO4]	3D Motion ICA [CLO3]
12	**Forced Vibration & Course Wrap-Up** By the end of this week, learners will be able to: • Determine the steady-state response of an undamped and damped 1 degree-of-freedom system [CLO4] • Apply dynamics and vibrations principles to finalize and defend a mechanism design [CLO4] [CLO5]	Final Project [CLO5], Midterm #2 [CLO3]

Note: The order of the delivery of the course is subject to change. Any changes will always be made in the students' favour to maximize learning. **The assessment due dates listed on the OnQ course page are the official due dates and take precedence over those listed in this syllabus.**

COURSE COMMUNICATION

Questions About Course Material

Questions or comments regarding the course material that can be of benefit to other students should be posted in the Q&A forum on the class website. The instructor, TAs, and students are encouraged to answer these questions directly in the discussion forum for the benefit of everyone in the course.

Course Announcements

The instructor will routinely post course news in the Announcements section on the main course homepage on OnQ. Please sign up to be automatically notified by email when the instructor posts new information in the Announcements section. Instructions on how to modify your notifications are found in the **Begin Here** section of the onQ course site.

Office Hours

You will have the opportunity to interact with either a TA or the instructor through Active Learning Sessions or by appointment office hours.

Confidential Matters

If you have a confidential matter you would like to discuss with your instructor, their contact details are on the first page of this document. Expect email replies within 48 hours.

Absences (Academic Considerations) and missed assignments

Students who miss assignments, quizzes, classes, labs, tutorials or other course deliverables due to unforeseen personal circumstances outside of their control can apply for Academic Consideration. For information on academic considerations due to extenuating circumstances, please review the information on the Smith Engineering website. Note that unacceptable reasons include extracurricular activities, work, travel plans, scheduling conflicts, generally being behind on schoolwork, etc. Do not schedule travel during midterms and final exams, as travel and work are not acceptable reasons for granting academic considerations.

If possible, missed Midterms will be made up by the student outside of class hours, within a week of the scheduled examination. If this is not possible, the weighting of the missed midterm will be split between the other midterm (+5%) and the final examination (+10%).

If students are to miss an Active Learning Session, they are responsible for reaching out to their group to contribute to the assignment. If the group notes in their submission that a student did not contribute to an assignment, that student will receive an opportunity to submit the assignment independently of the group for credit.

All questions around absences can be emailed to q.yetman@queensu.ca

Late Submission Policy

In the event of extenuating circumstances, you must follow the policies for requesting an academic consideration (as described above) or engage the appropriate Academic Accommodation process (see Academic Accommodation section below). In the absence of an approved consideration request, or accommodation in VENTUS, normal late penalty will apply (5% per day).

Requesting Academic Consideration Related to Group Work

Group submissions may pose challenges when one or more members of the group experience extenuating circumstances. The Active Learning Session Assignments and the Final Project are completed as a group.

Students should be able to complete the Active Learning Session assignments in the allotted Active Learning Session times, but to accommodate the needs of all learners, you have been given **3 days** to complete the group assignment. Students have **6 weeks** to complete the final project. As such, no further accommodation or extensions will be applied for the group assignments. Any late penalties will be listed in the group assignment description. Exceptions may be made as deemed required by the course instructor.

STANDARD QUEEN'S AND SMITH ENGINEERING POLICIES

Netiquette

In this course, you may be expected to communicate with your peers and the teaching team through electronic communication. You are expected to use the utmost respect in your dealings with your colleagues or when participating in activities, discussions, and online communication.

The following is a list of netiquette guidelines. Please read them carefully and use them to guide your online communication in this course and beyond.

1. Make a personal commitment to learn about, understand, and support your peers.
2. Assume the best of others and expect the best of them.
3. Acknowledge the impact of oppression on the lives of other people and make sure your writing is respectful and inclusive.
4. Recognize and value the experiences, abilities, and knowledge each person brings.
5. Pay close attention to what your peers write before you respond. Think through and re-read your writings before you post or send them to others.
6. It's alright to disagree with ideas, but do not make personal attacks.
7. Be open to be challenged or confronted on your ideas and challenge others with the intent of facilitating growth. Do not demean or embarrass others.
8. Encourage others to develop and share their ideas.

STUDENT CODE OF CONDUCT

Queen's University values maintaining an environment free of, and will not tolerate, harassment, discrimination, and reprisal. The Student Code of Conduct applies to all students at Queen's. It outlines the activities and behaviours that could be considered Non-Academic Misconduct (NAM). The Code also describes the NAM process and the sanctions that could be imposed on a student found responsible for a violation. All students should be familiar with the Student code of conduct and related policies on sexual violence prevention and response and harassment and discrimination prevention and response.
https://www.queensu.ca/nonacademicmisconduct/policies

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ACADEMIC INTEGRITY

As an engineering student, you have made a decision to join us in the profession of engineering, a long-respected profession with high standards of behaviour. As future engineers, we expect you to behave with integrity at all times. Please note that Engineers have a duty to:

- Act at all times with devotion to the high ideals of personal honour and professional integrity.
- Give proper credit for engineering work

The standard of behaviour expected of professional engineers is explained in the Professional Engineers Ontario Code of Ethics. Information on policies concerning academic integrity is available in the Queen's University Code of Conduct, in the Senate Academic Integrity Policy Statement, on the Smith Engineering website, and from your instructor.

Departures from academic integrity include plagiarism, use of unauthorized materials or services, facilitation, forgery, falsification, unauthorized use of intellectual property, and collaboration, and are antithetical to the development of an academic community at Queen's. Given the seriousness of these matters, actions which contravene the regulation on academic integrity carry sanctions that can range from a warning or the loss of grades on an assignment to the failure of a course to a requirement to withdraw from the University.

In the case of online or remotely proctored exams, impersonating another student, copying from another student, making information available to another student about the exam questions or possible answers, posting materials to online services, communicating with another person during an exam or about an exam during the exam window, or accessing unauthorized materials, including internet sources and using unauthorized materials, including smart devices, are actions in contravention of academic integrity.

Plagiarism/Use of Generative Artificial Intelligence (GenAI) Tools:

Students must submit their own work and cite the work that is not theirs. Any other use constitutes a Departure from Academic Integrity.

GenAI tools are permitted in this course under the following circumstances. Note that you may only use GenAI technologies to aid your learning, but not to replace it:

1. LibreChat and Copilot are the only GenAI tools recommended by Queen's University, other tools may be used at the student's discretion but are not recommended. Further information regarding AI Applications at Queen's can be found here.
2. AI tools are permissible only if explicitly noted in the activity and/or assignment instructions.
3. If any GenAI tools have been used, you must disclose the use and purpose and include proper citation of the generated materials. For information on how to cite the use of GenAI tools, please visit Queen's University Library.

Invalid Exams

An exam may be declared invalid in case of an interruption in an in-person examination; if the instructions in a remote or online exam were not followed; if the student uploads wrong materials; or if a situation arises where the integrity of the exam cannot be verified. If an exam is declared invalid, the student may be granted a re-write.

ACADEMIC AND STUDENT SUPPORT

Queen's has a robust set of supports available to you, including the Library, Student Academic Success Services (Learning Strategies and Writing Centre), and Career Services. Learners are encouraged to visit the Smith Engineering Current Students web portal for information about various other policies such as academic advisors, registration, student exchanges, awards and scholarships, etc. Students are also encouraged to review the information that is available in the EngQ Hub, posted in onQ.

Absences - Academic Considerations

Academic Consideration is a process by which students request validated time away from their academics when they are experiencing an extenuating circumstance or personal circumstances beyond a student's control that interfere with their ability to complete academic requirements.

For additional information and to apply for academic consideration, please review the information on the Smith Engineering's Academic Consideration (Absences) website.

Academic Accommodations

Academic Accommodations are available when students need an adjustment to their learning environment due to a reason related to a Human Rights protected ground. Below there is information on the most common Academic Accommodations requested by students. For more information, please see Smith Engineering's Academic Accommodation website.

Absences – Requests for Excused Absences for Significant Events (REASE)

Students participating in a Varsity Athletics event, the student reserve forces, a non-varsity event at the provincial, national, or international level, or an event to which you were invited as a distinguished guest can submit a Request for Excused Absence for Significant Event following the REASE process outlined on the Student Affairs webpage.

Students participating in Design Competitions supported by Smith Engineering can submit a Request for Excused Absence for Significant Events (REASE) following the process outlined on the Smith Engineering's Academic Consideration (Absences) website.

All Requests for Excused Absences for Significant Events must be submitted to the Faculty a minimum of 2 weeks prior to the event start date.

Religious Accommodations

Students in need of accommodation for religious observance are asked to complete the Religious Observance and Academics Activities form within a week of receiving their course syllabus (each term) and submit the form via email to the Engineering Program Advisor, (Accommodations & Considerations), engineering.aac@queensu.ca

- Alternative assignments are considered a "reasonable accommodation" under the Ontario Human Rights Code
- Please refer to the Queen's Multi-faith Calendar for all of this year's faith dates
- Students with questions about their rights and responsibilities regarding religious accommodations can contact chaplain@queensu.ca

Other human-rights based Accommodations

Engineering students who need accommodation on Human Rights grounds are asked to contact, in confidence, the Engineering Program Advisor, (Accommodations & Considerations) engineering.aac@queensu.ca or to call 613-533-6000 x 78013.

Accommodations for Disabilities

Queen's University is committed to working with students with disabilities to remove barriers to their academic goals. Queen's Student Accessibility Services (QSAS), students with disabilities, instructors, and faculty staff work together to provide and implement academic accommodations designed to allow students with disabilities equitable access to all course material (including in-class as well as exams). If you are a student currently experiencing barriers to your academics due to disability related reasons, and you would like to understand whether academic accommodations could support the removal of those barriers, please visit the QSAS website to learn more about academic accommodation.

To start the registration process with QSAS, click the "**Access Ventus**" button found on the Ventus student portal.

Ventus is an online portal that connects students, instructors, Queen's Student Accessibility Services, the Exam's Office, and other support services in the process to request, assess, and implement academic accommodations. To learn more about Ventus, visit A Visual Guide to Ventus for Students:
https://www.queensu.ca/ventus-support/students/visual-guide-ventus-students

For questions or assistance with requesting Academic Consideration or Accommodation, contact the Smith Engineering Program Advisor (Accommodations and Considerations) at engineering.aac@queensu.ca

Every effort has been made to provide course materials that are accessible. For further information on accessibility compliance of the educational technologies used in this course, please consult the links below.

Student Wellbeing

Success in Engineering includes taking care of your wellbeing. If you're experiencing stress, anxiety, burnout, personal challenges, or you're simply not sure where to turn, EngWell is here to help.

EngWell provides confidential support through our Wellness Advisor, Embedded Counsellors, mental health programming, and referrals to academic and campus resources. Whether you need short term guidance, help navigating university supports, or someone to talk to, we encourage you to reach out early.

Educational Technology

Educational Technology	Accessibility Compliance Information
onQ (Brightspace Learning Management System by D2L)	https://www.d2l.com/accessibility/standards/

If you find any element of this course difficult to access, please discuss with your instructor how you can obtain an accommodation.

Technical Support

Some basic comfort levels with basic hardware and software skills are required for this course. If you require technical assistance, please contact Technical Support.